

A

GSE208536 GeoMx cohort

Adjacent normal
(n = 16)

ADM
(n = 16)

PDAC
(n = 16)

48 spatial regions
verified by matching GeoMx
segment identifiers with
GEO sample metadata

B

Spatial validation

628
malignant-switch genes
from GSE143754

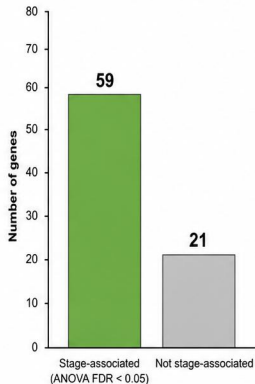
GeoMx expression
platform coverage

80 spatially validated
genes

Genes were retained only when
represented on the GeoMx
expression platform.

C

ANOVA stage-association results

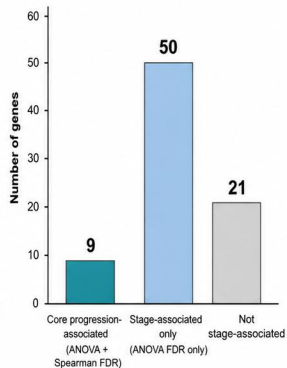


Benjamini-Hochberg correction
across 80 spatially validated genes.

D

Ordinal association classification

9 genes were significant in both ANOVA FDR
and ordinal Spearman FDR.



Classification of the 80 spatially validated
genes according to stage-association
evidence.